

6.5 Content Parsing & Preprocessing Implementation

This module handles normalization and preparation of uploaded content before it enters the embedding pipeline. Instead of focusing on extraction techniques, the system emphasizes structuring and cleaning parsed content so that it becomes suitable for reliable retrieval.

6.5.1 Preprocessing Techniques

Applied transformations include:

- Whitespace normalization
- Removal of formatting artifacts
- Sentence boundary correction
- Bullet and list standardization
- Table-to-text normalization for semi-structured documents
- Header and section-aware segmentation

These steps improve chunk quality and retrieval accuracy downstream.

6.6 Embedding + RAG Implementation

This layer forms the core intelligence of the system.

6.6.1 Chunking Logic

Chunks are generated with controlled overlap to preserve semantic continuity.

```
def chunk_text(text, size=800, overlap=100):  
    ...
```

6.6.2 Embedding Logic

Each chunk is transformed into a semantic vector.

```
vector = embedding_model.embed(chunk)
```

6.6.3 Retrieval Logic

The vector database performs similarity-based search.

```
vector_db.search(query_embedding)
```

This pipeline ensures that all responses remain grounded in retrieved internal knowledge.

6.7 Structured Data Utilization (Excel Processor)

YantraBuddy processes Excel sheets containing product and operational data and indexes each row into the unified semantic space.

Typical supported fields include:

- SKU
- Warranty terms
- Pricing
- Model number
- Compatibility notes

6.7.1 Excel → JSON Conversion Algorithm

```
for row in dataframe:
    entry = {
        "sku": row["SKU"],
        "warranty": row["Warranty"],
        "price": row["Price"],
        ...
    }
    embed_and_store(entry)
```

This approach allows structured enterprise data to become fully retrievable via natural language queries.

6.8 Frontend Implementation (Next.js)

6.8.1 Chat Interface Features

The frontend focuses heavily on operator usability.

Key features include:

- Fixed-height input box to prevent UI jumping
- Scrollable chat container for long conversations
- File upload support for knowledge ingestion
- Bilingual answer rendering (user language + English)
- Mobile-responsive layout for field usage

6.8.2 Frontend Code: